Your voice is no longer proof that it's you.

Scam Call Shield — real-time detection of phone scams and cloned voices, trained and evaluated on AMD GPUs via ROCm.

Vishal Kagade github.com/Vishalkagade/amd-hackthaon ■ human speech ■ synthesised

Three seconds of audio is enough to become someone else.

Voice cloning crossed the indistinguishable threshold, and the attack it enables — a panicked call from someone you love — defeats the one check people actually use: **recognising the voice.**

1,210%	**$3B+**	**1 in 4**	**$18k**
rise in AI-enabled scams during 2025	FBI-reported losses involving voice cloning	adults has already received an AI voice scam call	average loss for those who engage

Sources: FBI IC3, Europol (2025). Existing defences — call-backs, code words — fail precisely when the victim is under pressure, in real time.

A call can lie in two independent ways. So we check both.

What is being said

Whisper transcribes the call locally on-GPU; an open LLM served by Fireworks AI scores scam patterns — urgency, impersonation, secrecy, payment requests — and gives the callee plain-language advice.

`audio → whisper (local) → LLM → risk + advice`

Who is speaking

A wav2vec2 detector, fine-tuned on AMD via ROCm, scores every 3-second window for synthesis artefacts and drives a live authenticity meter.

`audio → mel + wav2vec2 → AI-voice probability`

A human reading a scam script trips only the first. **A clone of your daughter saying something ordinary trips only the second** — and that is the attack nobody else is watching for.

The benign clone is the one that should scare you.

DETECTOR OUTPUT — CLIPS NEVER SEEN IN TRAINING

AUDIO	WORDS SPOKEN	SCAM RISK	AI-VOICE	VERDICT
My real voice	ordinary conversation	5 / 100	0.018	HUMAN — clear
Clone of me	"Send €500, don't tell anyone"	95 / 100	0.974	SCAM + AI VOICE
Clone of me	"I'll be late for dinner"	5 / 100	0.974	AI VOICE — words are innocent

Row three is the whole argument. Nothing in the transcript is suspicious — **a text-only scam detector sees a completely normal call.** But it isn't me speaking.

EVALUATION — FOUR TIERS, NO LEAKAGE

Most deepfake numbers are inflated. These aren't.

WAV2VEC2 DETECTOR, TRAINED & EVALUATED ON AMD GFX1100 · EVERY TIER SPEAKER- AND ENGINE-DISJOINT

TEST TIER	WHAT IT MEASURES	N	DETECTED	FALSE ALARMS
1 · Unseen voices	New TTS voices, engine seen in training	231	100%	–
2/3 · Voice clones	XTTS-v2 clones of speakers never trained on	175	89.7%	–
4 · Real deepfakes	In-the-Wild: internet deepfakes of public figures	3,393	98.6%	0.0%

Tier 4 is the headline: **98.6% detection on 3,393 real-world deepfakes, with zero false alarms** across 9,868 genuine recordings of the same people. Trained only on studio TTS, this same model caught just **13–23%** of real clones — engine-disjoint generalisation is the problem everyone skips.

The detector called me a robot.

Late in the build, the model scored my own phone recording at **0.891 — AI**. Not a threshold problem: a **data** problem. Every human sample we had was studio speech or compressed internet audio. Every AI sample was clean and mp3-encoded. The model had quietly learned *"clean + compressed = synthetic"*.

- **Codec augmentation** on both classes — compression stops being a class cue.
- **Voice enrollment** — the owner's voice joins the human class, exactly as a shipping app would do at setup.

HELD-OUT CLIPS OF MY VOICE — NEVER TRAINED ON

MODEL	AI-PROB	VERDICT
Before	0.891	AI ✕
After	0.018	HUMAN ✓
Clone of me	0.974	AI ✓

Honest cost: removing the compression shortcut lost us one YouTube deepfake (0.79 → 0.002) — a *voice-conversion* attack, a family we have no training data for. We report the miss.

Trained and evaluated on AMD, ROCm end to end.


```
"gpu":   "AMD GPU (gfx1100)"
"vram_gb": 51.5
"rocm":  true
"hip":   "7.2.53211"
"torch": "2.9.1"

"itw_val_acc":       0.97
"owner_heldout_acc": 1.00
```


- **Zero code changes between vendors.** ROCm exposes AMD GPUs through the standard `torch.cuda` API — the same training script runs unmodified on either stack, and the log records which one it ran on.
- **The LLM axis is AMD too.** Scam analysis is served by Fireworks AI, running open models on AMD accelerators.
- **Reproducible.** `train_on_amd.ipynb` rebuilds the dataset, trains both detectors, and regenerates the four-tier table.

WHERE THIS GOES

The model is one artefact. The pipeline is the asset.

SHIPPING NOW

Speakerphone MVP

The mic hears both sides of the call. Works today, on any phone, with no OS permissions.

NEXT

On-device detector

Quantised to int8, the detector runs on a phone NPU. No call audio ever leaves the handset.

THE REAL UNLOCK

Carrier & OEM

A network operator screens the call before it rings — protecting the people who will never install an app.

Every cloning engine that ships makes today's checkpoint staler. What holds is the loop we built: **generate an attack set for any new engine → retrain → re-measure across four honest tiers.** That is what a handset maker or a regulator cannot easily rebuild — and what makes this defensible past demo day.